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1 Introduction

The application to be considered relies on a collection of movies. A sample of the data sets is illustrated by figure 1. The attributes of a movie are an identifier `idimdb`, the title, the release data, the runtime, the budget and revenue, a link to a poster, an overview and a rating.

idimdb	title	release	runtime	budget	revenue	poster	overview	rating
tt0029583	Snow White and the Seven Dwarfs	1937-12-21	83	1488423	184925486	https://image.tmdb.org/t/p/original/1P9eGGIT7eV7kA...	A beautiful girl, Snow White, takes refuge in the ...	7.6
tt0032910	Pinocchio	1940-02-23	88	2600000	84300000	https://image.tmdb.org/t/p/original/ynkEEi296ofs9k...	Lonely toymaker Geppetto has his wishes answered w...	7.4
tt0033563	Dumbo	1941-10-31	64	812000	1600000	https://image.tmdb.org/t/p/original/hKDdlisMtsU9J...	Dumbo is a baby elephant born with over-sized ears...	7.2
tt0034492	Bambi	1942-08-14	70	858000	267447150	https://image.tmdb.org/t/p/original/wV9e2y4myJ4KMF...	Bambi's tale unfolds from season to season as the ...	7.3
tt0042332	Cinderella	1950-02-22	74	2900000	263591415	https://image.tmdb.org/t/p/original/4nssBcQUBadCTB...	Cinderella has faith her dreams of a better life w...	7.3
tt0046183	Peter Pan	1953-02-05	77	4000000	87404651	https://image.tmdb.org/t/p/original/fJJOs1yrhKIZc...	Leaving the safety of their nursery behind, Wendy,...	7.3
tt0048280	Lady and the Tramp	1955-06-22	76	4000000	36359037	https://image.tmdb.org/t/p/original/wXKeGea4htM7Yz...	Lady, a golden cocker spaniel, meets up with a mon...	7.3
tt0053285	Sleeping Beauty	1959-02-17	75	6000000	51600000	https://image.tmdb.org/t/p/original/j2mTyUukcLwDle...	A beautiful princess born in a faraway kingdom is ...	7.2
tt0055254	One Hundred and One Dalmatians	1961-01-25	79	4000000	85000000	https://image.tmdb.org/t/p/original/9kiDisS1sVb5Ll...	When a litter of dalmatian puppies are abducted by...	7.3
tt0057546	The Sword in the Stone	1963-12-25	79	3000000	22182353	https://image.tmdb.org/t/p/original/7lyeeuhGAJSNX...	Wart is a young boy who aspires to be a knight's s...	7.2
tt0058331	Mary Poppins	1964-08-27	139	6000000	103082380	https://image.tmdb.org/t/p/original/yO8UyO86uNBq12...	A magical nanny employs music and adventure to hel...	7.8

Figure 1: Sample

2 Deployment

Docker pull command

```
docker pull cassandra
```

Server (first time)

```
docker network create bigdata
docker run --name my_cassandra --network bigdata -d cassandra
```

Server (if already exists)

```
docker start my_cassandra
```

CQH Shell

```
docker exec -it my_cassandra cqlsh
```

or

```
docker exec -it my_cassandra bash
cqlsh
```

3 Import data

```
docker cp movies.csv my_cassandra:/tmp/movies.csv
```

4 Tutorial

4.1 Index creation

Create keyspace

```
CREATE KEYSPACE IF NOT EXISTS movies WITH REPLICATION = {'class' : 'SimpleStrategy',  
  'replication_factor': 1};  
USE movies;
```

Create table

```
DROP TABLE IF EXISTS movie;  
CREATE TABLE movie(  
  idimdb text,  
  title text,  
  release text,  
  runtime int,  
  budget int,  
  revenue int,  
  poster text,  
  rating float,  
  PRIMARY KEY (idimdb)  
  --PRIMARY KEY (title, release)  
);
```

Import data

```
COPY movie (idimdb, title, release, runtime, budget, revenue, poster, rating) FROM '/tmp/movies.csv'  
  WITH DELIMITER=',';
```

5 Activities on querying and data models

5.1 Queries

Write the following queries:

1. All movies
2. Title and runtime of movies
3. 81 minutes movies
4. Movies which runtime is between 60 minutes and 80 minutes
5. Number of movies
6. Number of movies per runtime
7. Average budget
8. Most expensive movie
9. Titles of movies ordered in decreasing order by runtime
10. Title of three movies that are longer than 140 minutes

5.2 Models

Propose the following models:

1. Genders with a set of movies that fit within the gender
2. Movies with a set of actors
3. Studios with a set of movies that have been produced
4. Movies with all the elements (actors, studios and genders)

6 Activities on consistent hashing

In Cassandra, consistent hashing makes it possible to distribute data across a data center/cluster and to determine the placement of copies.

6.1 Virtual 2-node cluster with replication factor 1

6.1.1 Deployment

Edit a file docker-compose.yml.

```
version: '3'
services:
  cassandra-node1:
    image: cassandra:latest
    container_name: cassandra-node1
    ports:
      - "9042:9042"
    environment:
      CASSANDRA_CLUSTER_NAME: "TestCluster"
      CASSANDRA_SEEDS: "cassandra-node1"
      CASSANDRA_BROADCAST_ADDRESS: "cassandra-node1"
      CASSANDRA_ENDPOINT_SNITCH: GossipingPropertyFileSnitch

  cassandra-node2:
    image: cassandra:latest
    container_name: cassandra-node2
    environment:
      CASSANDRA_CLUSTER_NAME: "TestCluster"
      CASSANDRA_SEEDS: "cassandra-node1"
      CASSANDRA_BROADCAST_ADDRESS: "cassandra-node2"
      CASSANDRA_ENDPOINT_SNITCH: GossipingPropertyFileSnitch
```

Start the cluster.

```
docker-compose up -d
```

CQL shell.

```
docker exec -it cassandra-node1 cqlsh
```

Create the environment with the keyspace univ and the table user.

The following command is using docker inspect with a Go template (-f = format) to extract the container's IP address inside its Docker network(s).

```
docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' cassandra-node2
```

6.1.2 Exercise

Insert in the table user the following records:

```
1,Alice,alice@example.com
2,Bob,bob@example.com
3,Charlie,charlie@example.com'
```

Execute the query to get all the users.

Switch off cassandra-node2.

Execute the query to get all the users.

Provide a brief explanation.

Switch on cassandra-node2.

Wait a little and execute the query to get all the users.

It has to be noted that the following command makes it possible to know within the keyspace ks1 the nodes storing the record of the table t1 with value 'v1' on attribute a1 used as a primary key (for hashing).

```
nodetool getendpoints ks1 t1 'v1'
```

Give the node storing the different records.

It has to be noted that the following command enables to get the IP of a node 'my-node' on the Docker cluster.

```
docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' my-node
```

Give the absent record(s) when cassandra-node2 is off-line.

6.2 Virtual 2-node cluster with replication factor 2

6.2.1 Exercise

Delete the keyspace univ.

Create the environment with the keyspace univ and the table user.

Insert in the table user the following records:

```
1,Alice,alice@example.com
2,Bob,bob@example.com
3,Charlie,charlie@example.com'
```

Execute the query to get all the users.

Switch off cassandra-node2.

Execute the query to get all the users.

Provide a brief explanation.

Check with nodetool.

Switch on cassandra-node2.

6.3 Virtual 3-node cluster with replication factor 3

6.3.1 Exercise

Propose the global process to deploy a 3-node cluster with a replication factor 3 and propose some test to observe the data management.

7 References

- **Lecture:** https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/big_data_cassandra.pdf
- **Cassandra:** https://cassandra.apache.org/_/index.html